

Jaewon Yoon

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EDUCATION

Seoul National University

03 2021 — 02 2027

B.S. in Biological Sciences, Interdisciplinary Major of Artificial Intelligence - (GPA - 3.88/4.3)

COURSEWORK / SKILLS

Cell Biology	Genetics	Biochemistry 1,2
Systems Biology	Immunology	Introduction to Computer Science for Biologists
Introduction to Bioinformatics	Bioinformatics and Practice 2	Virology
Computer Programming	Data Structure	Algorithm
Linear Algebra 1	Concepts and Applications of Probability	
Introduction to Data Mining	Machine Learning Fundamentals	

RESEARCH EXPERIENCE

SNU Laboratory of Computational Biology

12 2024 - 02 2025

Undergraduate Researcher

- Explored the applications of Deep - Learning approaches to transcriptomic data.
- Developed a deep learning model to predict RNA modifications from sequencing data produced by Nanopore sequencing.

SNU Steinegger Lab (Laboratory of Machine Learning and Bioinformatics)

07 2025 - Present

Undergraduate Researcher

- Developed methods for computing significance scores for matches of FoldDisco (a software for protein motif search)
- Currently developing a hybrid protein structure predicting model, combining architectures of different versions of AlphaFold

PUBLICATIONS

Journal paper

- Kim, H., Kim, R.S., Mirdita, M., **Yoon, J.**, Steinegger, M. (2026). Structural motif search across the protein-universe with FoldDisco *Nature Biotechnology*, doi: 10.1038/s41587-026-03162-9
- **Yoon, J.***, Kim, H.*, Steinegger, M. (2026). Improving structural sorting outputs of FoldDisco *Biorxiv*, doi: TBD
- Lueth, F.*, **Yoon, J.***, Kim, Y., Steinegger, M. (2026). FrankenFold: Integrating structure modules to AlphaFold3 resolves hallucination issues caused by diffusion models *Biorxiv*, doi: TBD

Others

- Nam, J., Woo, J., Yoon, J. et al., (2025). Unfolding Neuroscience, to be published online.

SCHOLARSHIP / AWARDS

Presidential Science Scholarship — Korea Student Aid Foundation

2021-2026

- 150 undergraduate students selected nationwide for excellency

Encouragement Prize — SNU College, Seoul National University

02 2026

- Awarded to outstanding class founders of the 2025 Student-Directed Education Programs

Dean's list — College of Natural Sciences, Seoul National University

Autumn 2021, Autumn 2025

- Awarded to outstanding undergraduate students each semester

Army Commendation Medal — Eighth Army, United States of America

08 2023

- Awarded to winners of General Paik Sun-Yup Leadership Board hosted by the Eighth Army, USFK

Korea Talented Award — Ministry of Education, ROK	11 2020
• 50 high school students and 50 university students selected nationwide	
International Biology Olympiad Silver Medal	08 2020
• Scientific competition between international teams of secondary school students	

EXTRACURRICULAR - ACADEMIC

Teaching assistant, SNU college	06 2026 - 08 2026
<i>Undergraduate student tutor</i>	

- Assisted teaching of the course Basic Computing: First Adventures in Computing
- Assisted and advised undergraduate students in programming exercises and projects

Regional Student Group - Korea Core Team, ISCB	01 2026 - Present
<i>Member, Program Committee</i>	

- Planned and organized annual symposiums, workshops, and web seminars for RSG-Korea of ISCB
- Invited multiple scholars focusing on antibody structure prediction or omics data analysis for seminars

Deep-learning-based bioinformatics seminar - student-directed education seminar	Autumn 2025
<i>Class founder</i>	

- Held weekly student seminars on investigating various fields and researches of computational biology
- Invited multiple scholars focusing on protein structure prediction, protein - ligand interaction prediction, and genomic data analysis for lecture
- Organized presentations and discussions regarding deep-learning based bioinformatics research

Synapse Scholars	07 2024 - Present
<i>Deputy Manager</i>	

- Participated in drafting, writing, and publishing the online neurobiology textbook, *Unfolding Neuroscience*
- Wrote or co-wrote the chapters Sensation, Immunology, and Artificial Intelligence

EXTRACURRICULAR - NON-ACADEMIC

SNU College of Natural Sciences Undergraduate Student Mentor (Multi-major)	2025, 2026
<i>Undergraduate Student Mentor</i>	

- Provided advice to undergraduate students on selecting and preparing various double majors.

SNU CNS Science Volunteering Group	2022, 2024-2026
<i>Group member / Member of Group task force - mentoring division</i>	

- Participated in mentoring and advising high school students of rural areas nationwide.
- Established and directed multiple programs to academically advise students.

SNU School of Biological Sciences Undergraduate Student Mentor	03 2025 - 02 2026
<i>Undergraduate Student Mentor</i>	

- Advised freshman and sophomore students of the department as an undergraduate student mentor.
- Aided overall school life and provided counseling regarding career paths for fellow students.

MSC-K HQ, 19th ESC, 8th Army, USFK (Military Service)	08 2022 - 02 2024
<i>KATUSA soldier / Discharged as army sergeant (E-5)</i>	

- Served as a Korean soldier augmented to the USFK for mandatory military service of ROK.
- Supported communication and cooperation between USFK stationed in Camp Carroll and Korean army counterparts.

REFERENCES

Prof. Martin Steinegger

Associate Professor, School of Biological Sciences, Seoul National University

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Scholar Profiles: SteineggerLab Homepage — Seoul National University - Personal Page

Prof. Minkyung Baek

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Prof. Milot Mirdita

Assistant Professor, Department of Precision Medicine, Seungkyunkwan University

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